

CS-GY 6643 Computer Vision – Project 1, Kaggle Competition Report

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Kaggle team name: (exactly as it appears on the leaderboard)

1 Introduction and background

State the competition task precisely: what the input is, what you are asked to predict, and how submissions are scored. Explain what makes the task difficult from a computer vision standpoint, for example the degradations present in the data, the amount of labelled data available, or the gap between the public and private splits. Summarise how this class of problem has been approached before, in the lectures and in the literature, and cite your sources (Canny 1986; Lowe 2004). Close by outlining the approach you took and why you expected it to work on this data.

2 Datasets

2.1 Preprocessing

Describe the competition data and every preprocessing step you applied, in the order you applied it: resizing, normalisation, colour conversion, denoising, augmentation, and how you constructed your training and validation splits. Include figures showing what the data actually looks like, the kind of noise or degradation you observed, and any class imbalance or missing or corrupted samples. State how your pipeline handles each of these.

Include a figure showing the data and the effect of your preprocessing: representative samples, the degradations or artefacts you observed, the label distribution, and one example shown before and after each preprocessing stage. Write a caption that can be read on its own, without the body text.

2.2 External and Auxiliary Datasets

Declare any external data, pretrained weights, or auxiliary datasets you used, with a link or citation for each. If you used none, say so explicitly.

3 Methods

Give the formulation of the problem: define your inputs, outputs, and the loss or objective you optimise. Describe your pipeline end to end, using a diagram where it is clearer than prose.

3.1 Proposed Method

Include a diagram of your architecture or pipeline, showing the inputs, each processing stage, and the output, and explain the reasoning behind each design choice. If you tried alternatives before settling on this one, say what they were and why you moved on. Reference any relevant literature or lecture material that the method builds on.

4 Results

4.1 Training Details

Outline your training and validation protocol: the split you used and why, the hyperparameters you searched and the ranges searched over, and the final values selected. Report the random seeds, the hardware, and the total runtime, so that somebody else could reproduce your numbers from your submitted code.

4.2 Results

Report your validation results alongside your public and private leaderboard scores, and account for any gap between them. Include the competition metric and any secondary metrics that support your conclusions, with standard deviations across seeds or folds where relevant. Include an ablation showing what each major component of your method contributes.

5 Use of AI

If you used any AI-assisted tools while completing this project, you must clearly disclose their use. Please include:

- The AI tool or model used (e.g., ChatGPT, Claude, Gemini, GitHub Copilot, etc.).
- The exact LLM prompts used.
- Your rationale for using AI for those specific tasks.
- The resulting output and a brief explanation of how it was incorporated into your work.

If you did **not** use any AI-assisted tools, explicitly state that no AI was used in completing the project.

Important: If a submission claims that no AI was used but evidence of AI-assisted work is identified during evaluation, the discrepancy will be treated seriously and may result in strict action in accordance with the NYU's academic integrity policies.

6 Conclusions

What did you learn from your experiments? What are existing limitations and ongoing issues? What changes would you make going forward?

7 Author contributions

You must explicitly state who in the team worked on which parts of the project.

8 References

Please make sure to include references for what you cite. **LLMs often get these wrong, so please double check if you use LLMs here!**

References

- Canny, J. (1986). "A computational approach to edge detection". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 8.6, pp. 679–698.
- Lowe, D. G. (2004). "Distinctive image features from scale-invariant keypoints". In: *International Journal of Computer Vision* 60.2, pp. 91–110.